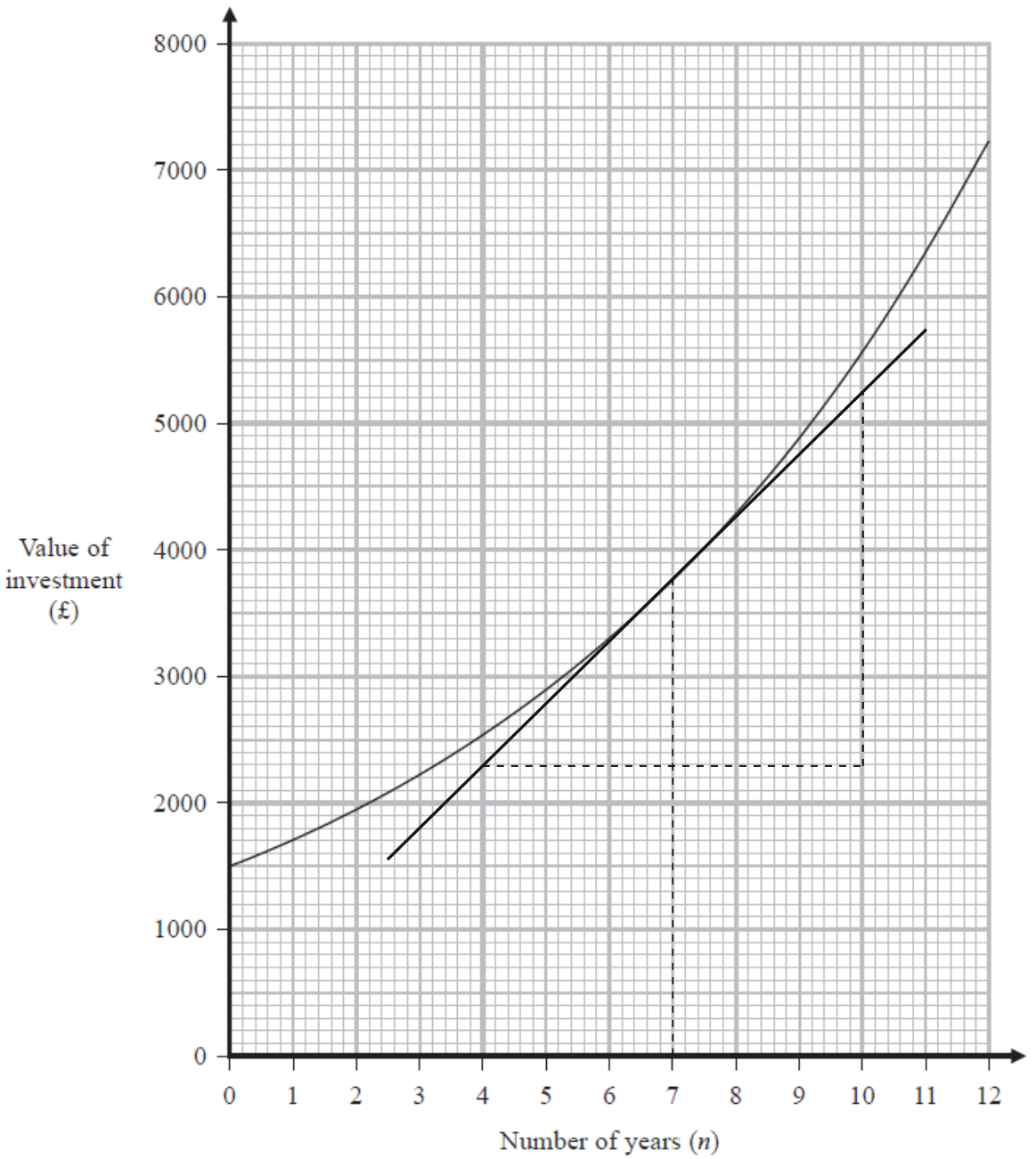
18 Frank invested an amount of money for 12 years.

The graph shows the value of Frank's investment over the 12 year period.



(a) (i) Write down the amount of money that Frank invested.

At n = 0, value = £1500.

1 mark
Read y-intercept.

£.....

(1)

Frank said that 12 years after he had made his investment, it was worth more than five times its original value.

(ii) Was Frank correct?
Give a reason for your answer.

No: 5 × £1500 = £7500.

At 12 years the graph is about £7250, below £7500.

1 mark
Compare with five times the original.

(1)

(b) (i) Find an estimate for the gradient of the curve at $n = 7$

Use the tangent drawn on the original graph.

1 mark
Tangent at n = 7.

Two readings: (4 years, £2290), (10 years, £5250).

Gradient ≈ (5250 − 2290) ÷ (10 − 4)

1 mark
Rise ÷ run.

= 2960 ÷ 6 = 493.33...

Estimate ≈ £490 per year

1 mark
About £490/year.

(3)

(ii) Explain what this gradient represents.

At year 7, the investment is increasing in value at about £490 per year.

1 mark
Growth rate at year 7.

(1)

(Total for Question 18 is 6 marks)